

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\144-D.pvdb

Date of Analysis : May 7, 2026 5:58pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3008.000 mm OD	
Vessel Design Length, Tangent to Tangent	312.50 cm
Distance of Bottom Tangent above Grade	400.00 cm
Distance of Base above Grade	30.00 cm
Specified Datum Line Distance	370.00 cm
Internal Design Temperature	100 °C
Internal Design Pressure	46.405 kgf/cm ²
External Design Temperature	0 °C
Maximum Allowable Working Pressure	47 kgf/cm ²
Shop Test Pressure	60.327 kgf/cm ²
Field Test Pressure	60.327 kgf/cm ²
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-12.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Basing	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt 2	...	...	...	1.6000	N/A	No
Skirt 1	...	...	...	1.6000	N/A	No
Bottom Head	46.553	0.00	47.000	3.0000	No	No
Shell-1	46.475	0.00	47.000	3.0000	N/A	No
Top Head	46.405	0.00	47.000	3.0000	No	No

Liquid Level: 147.50 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-100.0	270.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	100.0	14.0	14.0	...	...	0.70	0.50
Ellipse	5.0	5.0	60.0	52.0	51.1	...	1.00	1.00
Cylinder	307.5	302.5	53.0	53.0	52.0	...	1.00	1.00
Ellipse	312.5	5.0	60.0	52.0	50.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_3"	WRC-107/537	0.882	n/a	Passed
N1_24"	WRC-297	0.976	Shell	Passed
N4_1.5"	WRC-297	0.976	Shell	Passed
N2_24"	WRC-107/537	0.880	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	135	2136.13	10968.4	10095.1	64470
20	30	320	1734.98	10789.4	4869.11	35097
30	40	372.5	1586.41	10686.5	3208.41	24359.1
40	50	526.25	1183.69	9195.75	3139.16	23862
50	60	680	716.556	5287.72	265.034	1955.78

Abs Max of the all of the Stress Ratio's : 0.4871

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm
Inside Diameter of Basing	2791.0000	mm
Outside Diameter of Basing	3251.0000	mm
Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	3111.0002	mm
Number of Bolts	16	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	130.0000	mm
Height of Gussets	230.0000	mm
Distance between Gussets	90.0000	mm
Thickness of Top Plate or Ring	38.0000	mm
Radial Width of the Top Plate	140.0000	mm

Basing Sketch for Cylindrical Skirts:

Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	6893.	kgf
Total Earthquake Shear on Support	15726.	kgf
Wind Moment on Support	86970.	kgf-m.
Earthquake Moment on Support	141345.	kgf-m.
Maximum Initial Bolt Load	6808.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	37037.0	kg
Shop Test - Fabricated + water (full)	64223.7	kg
Shipping - Fab. + rem. intls.+ shipping app.	46657.7	kg
Erected - Fab. + rem. intls.+ insul. (etc)	46802.7	kg
Empty - Fab. + intls. + details + wgths.	46802.7	kg
Operating - Empty + operating liquid (no ca)	69951.5	kg
Field Test - Empty weight + test medium (full)	73844.4	kg



Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_3"	47	...	...	OK	7.80	OK	Passed
N1_24"	47	...	...	OK	12.92	OK	Passed
N4_1.5"	47	...	...	OK	7.52	OK	No Calc[*]
K1_2"	47	...	...	OK	7.80	OK	No Calc[*]
K2_2"	47	...	...	OK	7.80	OK	No Calc[*]
K3_2"	47	...	...	OK	7.80	OK	No Calc[*]
K4_2"	47	...	...	OK	7.80	OK	No Calc[*]
M1_24"	47	...	...	...		OK	Passed
N2_24"	47	...	...	OK	12.92	OK	Passed
K5_2"	47	...	...	OK	6.42	OK	No Calc[*]
K6_2"	47	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 47 Nozzle : M124"
 Minimum MAWP Shells/Flanges : 47 Element : Bottom Head
 Minimum MAPnc Shells/Flanges : 50 Element : Top Head

Computed Vessel M.A.W.P. : 47 kgf/cm²

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3_3"	0.000	0.000	174.200
40	N1_24"	1765.000	0.000	1175.200
40	N4_1.5"	1400.000	180.000	170.104
40	K1_2"	1080.000	0.000	184.328
40	K2_2"	150.000	0.000	184.328
40	K3_2"	850.000	180.000	184.328
40	K4_2"	150.000	180.000	184.328
40	M1_24"	1000.000	270.000	1175.200
50	N2_24"	0.000	0.000	1175.200
50	K5_2"	0.000	0.000	148.800
50	K6_2"	0.000	180.000	148.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	353.42	900
K1_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
K2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
K3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
K4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
K5_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	366.55	300
K6_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	366.55	300
N3_3"	3.000 in	Actual	WNF	3.000	10.000	...	...	202.27	600
N1_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	3213.00	300
M1_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	385.37	300
N2_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	569.58	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4_1.5"	SA-350 LF2	52.000	10.000	...	...	...
K1_2"	SA-350 LF2	52.000	10.000	...	...	...
K2_2"	SA-350 LF2	52.000	10.000	...	...	...
K3_2"	SA-350 LF2	52.000	10.000	...	...	...
K4_2"	SA-350 LF2	52.000	10.000	...	...	...
K5_2"	SA-350 LF2	52.000	10.000	...	...	...
K6_2"	SA-350 LF2	52.000	10.000	...	...	...
N3_3"	SA-350 LF2	52.000	10.000	...	...	...
N1_24"	SA-350 LF2	52.000	10.000	...	...	14.000
M1_24"	SA-350 LF2	52.000	10.000	...	...	...
N2_24"	SA-350 LF2	52.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4_1.5"	140.000	180.0	300.00	0.00	Shell-1
K1_2"	108.000	0.0	300.00	0.00	Shell-1
K2_2"	15.000	0.0	300.00	0.00	Shell-1
K3_2"	85.000	180.0	300.00	0.00	Shell-1
K4_2"	15.000	180.0	300.00	0.00	Shell-1
K5_2"	...	0.0	300.00	0.00	Top Head
K6_2"	...	180.0	300.00	0.00	Top Head
N3_3"	...	0.0	150.00	0.00	Bottom Head
N1_24"	176.500	0.0	300.00	2860.00	Shell-1
M1_24"	100.000	270.0	300.00	0.00	Shell-1
N2_24"	...	0.0	500.00	0.00	Top Head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 60.0mm THK X 2900.0mm ID X 5.0cm	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
2	LADDER: 4000.0mm	...
2	INSULATION: 775mm X 40mm THK	cold
1	INSULATION: 3025mm X 40mm THK	cold
1	PLATFORM: 1000mm X 5000mm WIDE	
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	BASERING: 3251.0mm OD X 2791.0 ID X 32.0mm	SA-516 70
16	BASERING BOLTS: 42.0mm DIA	SA-193 B7
16	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 3251.0mm OD X 2971.0 ID X 38.0mm	...
32	BASERING GUSSET PLATES: 130.0mm X 130.0mm X 230.0mm X 16.0mm	...
2	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3023.3mm X 270.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 1511.6mm X 270.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 1687.3mm X 270.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3081.9mm X 100.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3107.5mm X 100.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3056.2mm X 100.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 53.0mm THK X 3038.9mm X 302.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 53.0mm THK X 3064.2mm X 302.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 53.0mm THK X 3013.6mm X 302.5cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Bottom Head	[10]	!	-12	-12	0.992	52.000	1.00	Yes
Bottom Head	[7]	!	-12	-19	0.869	60.000	1.00	Yes
Shell-1	[8]	!	-12	-13	0.991	53.000	1.00	Yes
Top Head	[10]	!	-12	-13	0.989	52.000	1.00	Yes
Top Head	[7]	!	-12	-19	0.867	60.000	1.00	Yes
N33"	[1]	!	-46	-104	0.178	10.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-86				
N124"	[1]	!	-46	-46	0.902	14.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-52				
N41.5"	[1]	!	-46	-104	0.057	16.002	1.00	No
Nozzle Flg	[5]	!	-46	-46				
K12"	[1]	!	-46	-104	0.069	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46				
K22"	[1]	!	-46	-104	0.070	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46				
K32"	[1]	!	-46	-104	0.069	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46				
K42"	[1]	!	-46	-104	0.070	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46				
M124"	[1]	!	-46	-46	0.902	14.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-52				
N224"	[1]	!	-46	-46	0.902	14.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-52				
K52"	[1]	!	-46	-104	0.118	10.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-52				
K62"	[1]	!	-46	-104	0.118	10.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-52				
Warmest MDMT:		-12	-12					
Required Minimum Design Metal Temperature					-12.0	°C		
Warmest Computed Minimum Design Metal Temperature					-12.0	°C		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	46.405	kgf/cm ²
Design Internal Temperature	100.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 cm
Distance Upwind of Crest	Lh 0 cm
Distance from Crest to the Vessel	x 0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt 2	
Distance "FROM" to "TO"	270	cm
Skirt Outside Diameter	2971	mm
Diameter of Skirt at Base	2971	mm
Skirt Thickness	14	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	100	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Allowable Stress, Ambient	1407.2	kgf/cm ²
Allowable Stress, Operating	1407.2	kgf/cm ²
Allowable Stress, Hydrotest	1829.3	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2437	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing+Bracing	
Dist. from "FROM" Node / Offset dist	30	cm
Miscellaneous Weight	399.99	kgf
Offset from Element Centerline	1400	mm

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt 1	
Distance "FROM" to "TO"	100	cm
Skirt Outside Diameter	2971	mm
Diameter of Skirt at Base	2971	mm
Skirt Thickness	14	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	100	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	Bottom Head	
Distance "FROM" to "TO"	5	cm
Inside Diameter	2900	mm
Element Thickness	52	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	46.405	kgf/cm ²
Design Temperature Internal Pressure	100	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Platform	
Detail ID	BTL PLT	
Dist. from "FROM" Node / Offset dist	0	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	46256	cm ²
Platform Weight	3929.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	105	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	-72.5	cm
Height/Length of Liquid	77.5	cm
Liquid Density	999.54	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	-72.5	cm
Height/Length of Insulation	77.5	cm
Thickness of Insulation	40	mm
Density	0.00032	kg/cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N33"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	69.221	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	30	
Detail Type	Weight	
Detail ID	Anti-Swirl Baff	
Dist. from "FROM" Node / Offset dist	-40	cm
Miscellaneous Weight	9.9998	kgf
Offset from Element Centerline	0	mm



Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	302.5 cm
Element Outside Diameter	3008 mm
Element Thickness	53 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	46.405 kgf/cm ²
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	70 cm
Liquid Density	999.54 kg/m ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	302.5 cm
Thickness of Insulation	40 mm
Density	0.00032 kg/cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N124"
Dist. from "FROM" Node / Offset dist	171.5 cm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	N41.5"
Dist. from "FROM" Node / Offset dist	135 cm
Nozzle Diameter	1.5 in.
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K12"
Dist. from "FROM" Node / Offset dist	103 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf



Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K22"	
Dist. from "FROM" Node / Offset dist	10	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K32"	
Dist. from "FROM" Node / Offset dist	80	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K42"	
Dist. from "FROM" Node / Offset dist	10	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M124"	
Dist. from "FROM" Node / Offset dist	95	cm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	40	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	95	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	1800	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	277.5	cm
Miscellaneous Weight	145	kgf
Offset from Element Centerline	0	mm
Element From Node	40	



Detail Type	Weight	
Detail ID	Pipe Support	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister Top Sp	
Dist. from "FROM" Node / Offset dist	283	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister bottom	
Dist. from "FROM" Node / Offset dist	272	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Lifting Lug1	
Dist. from "FROM" Node / Offset dist	260	cm
Miscellaneous Weight	399.99	kgf
Offset from Element Centerline	1500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Lifting Lug2	
Dist. from "FROM" Node / Offset dist	260	cm
Miscellaneous Weight	399.99	kgf
Offset from Element Centerline	1500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	151.5	cm
Miscellaneous Weight	949.98	kgf
Offset from Element Centerline	0	mm

Element#	5/5	
Element From Node	50	
Element To Node	60	
Element Type	Elliptical	
Description	Top Head	
Distance "FROM" to "TO"	5	cm
Inside Diameter	2900	mm
Element Thickness	52	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	46.405	kgf/cm ²
Design Temperature Internal Pressure	100	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PTL-2	
Dist. from "FROM" Node / Offset dist	90	cm
Platform Start Angle (degrees)	0.0	



Platform End Angle (degrees)	0.0	
Platform Wind Area	66256	cm ²
Platform Weight	4403	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	5000	mm
Platform Height	1000	mm
Platform Clearance or End Offset	2500	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	5000	mm
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	77.5	cm
Thickness of Insulation	40	mm
Density	0.00032	kg/cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N224"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K52"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K62"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	50	
Detail Type	Weight	
Detail ID	PSV Weight	
Dist. from "FROM" Node / Offset dist	77.5	cm
Miscellaneous Weight	15000	kgf
Offset from Element Centerline	2200	mm
Element From Node	50	



Detail Type	For./Mom.
Detail ID	N2 F&M
Dist. from "FROM" Node / Offset dist	65 cm
Force in X Direction	3363.9 kgf
Force in Y Direction	-5045.8 kgf
Force in Z Direction	3363.9 kgf
Moment about X Axis	5651.4 kgf-m.
Moment about Y Axis	0 kgf-m.
Moment about Z Axis	5651.4 kgf-m.
Force/Moment Combination Method	SRSS

**XY Coordinate Calculations:**

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
	Skirt 2	...	270	...	270
	Skirt 1	...	370	...	100
	Bottom Head	...	375	...	5
	Shell-1	...	677.5	...	302.5
	Top Head	...	682.5	...	5

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Skirt 2	...	...	14	1.6	2971	...	1
Skirt 1	...	...	14	1.6	2971	...	2
Bottom Head		46.553	60	3	2900	1407.2	3
Shell-1		46.475	53	3	3008	1407.2	4
Top Head		46.405	60	3	2900	1407.2	5

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Skirt 2	...	...	No Calc	No Calc	14	No Calc	1
Skirt 1	...	...	No Calc	No Calc	14	No Calc	2
Bottom Head		46.4053	47.2769	50.2833	52	51.0965	3
Shell-1		46.4053	47.3411	50.2964	53	52.026	4
Top Head		46.4053	47.4243	50.2833	52	50.9436	5

Minimum

47

50

MAWP: 47 kgf/cm², limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Bottom Head

Elliptical Head Design Information:					
Design Pressure	P	46.553 kgf/cm²	Int. Design Temperature	100 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		5.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.2 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1381.3 kgf/cm²	
Spec. Min. Thk t _e , or s _a		52.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		103607.6 cm²	SG of Contents	1.000	
Element Volume		3522781.0 cm³	Weight of Contents	3521.2 kgf	
Empty Weight		5092.3 kgf	Operating Weight	8613.5 kgf	
			Comp. All. Ope. Stress	1428.9 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (46.553 \cdot 2906.0 \cdot 0.997) / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.553)$$

$$= 48.0965 + 3.0000 = 51.0965 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.147 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 49.0) / (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)$$

$$= 47 - 0 = 47 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 52.0) / (1.0 \cdot 2900.0 + 0.2 \cdot 52.0)$$

$$= 50 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (46.553 \cdot (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)) / (2 \cdot 1.0 \cdot 49.0)$$

$$= 1381.298 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (46.553 \cdot 1453.0) / (1407.16 \cdot 1.0 - 0.6 \cdot 46.553) + 3.0$$

$$= 52.043 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.075 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.16 \cdot 1.0 \cdot 57.0) / (1453.0 + 0.6 \cdot 57.0)$$

$$= 54 - 0.1 = 54 \text{ kgf/cm}^2$$



Factor K, corroded condition [Kcor]:
= (2 + (Inside Diameter/(2 • Inside Head Depth))²)/6
= (2 + (2906.0/(2 • 728.0))²)/6
= 0.997256

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 8.604 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	46.475 kgf/cm²	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		302.500 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.2 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.2 kgf/cm²
Outside Diameter	Do	3008.000 mm	Actual Stress		1379.4 kgf/cm²
Spec. Min. Thk t _e , or s _a		53.0000 mm	Spec. Nominal Thk.		53.000 mm
Surface Area		285859.8 cm²	SG of Contents		1.000
Element Volume		20008346.0 cm³	Weight of Contents		4627.9 kgf
Empty Weight		11534.8 kgf	Operating Weight		16162.7 kgf
			Comp. All. Ope. Stress		1432.1 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (46.475 \cdot 1504.0) / (1407.16 \cdot 1.0 + 0.4 \cdot 46.475)$
 $= 49.0260 + 3.0000 = 52.0260 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $\text{Less Operating Hydrostatic Head Pressure of } 0.070 \text{ kgf/cm}^2$
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1407.16 \cdot 1.0 \cdot 50.0) / (1504.0 - 0.4 \cdot 50.0)$
 $= 47 - 0.1 = 47 \text{ kgf/cm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (1407.16 \cdot 1.0 \cdot 53.0) / (1504.0 - 0.4 \cdot 53.0)$
 $= 50 \text{ kgf/cm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$
 $= (46.475 \cdot (1504.0 - 0.4 \cdot 50.0)) / (1.0 \cdot 50.0)$
 $= 1379.386 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / Ro))$ 1.794 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Top Head

Elliptical Head Design Information:					
Design Pressure	P	46.405 kgf/cm ²	Int. Design Temperature		100 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		5.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.2 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.2 kgf/cm ²
Inside Diameter	Di	2900.000 mm	Actual Stress		1376.9 kgf/cm ²
Spec. Min. Thk t _e , or s _a		52.0000 mm	Spec. Nominal Thk.		60.000 mm
Surface Area		103607.6 cm ²	SG of Contents		0.000
Element Volume		3522781.0 cm ³	Weight of Contents		0.0 kgf
Empty Weight		5092.3 kgf	Operating Weight		5092.3 kgf
			Comp. All. Ope. Stress		1428.9 kgf/cm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (46.405 \cdot 2906.0 \cdot 0.997) / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.405)$$

$$= 47.9436 + 3.0000 = 50.9436 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 49.0) / (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)$$

$$= 47 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.16 \cdot 1.0 \cdot 52.0) / (1.0 \cdot 2900.0 + 0.2 \cdot 52.0)$$

$$= 50 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (46.405 \cdot (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)) / (2 \cdot 1.0 \cdot 49.0)$$

$$= 1376.924 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (46.405 \cdot 1453.0) / (1407.16 \cdot 1.0 - 0.6 \cdot 46.405) + 3.0$$

$$= 51.884 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.16 \cdot 1.0 \cdot 57.0) / (1453.0 + 0.6 \cdot 57.0)$$

$$= 54 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (2906.0 / (2 \cdot 728.0))^2) / 6$$
$$= 0.997256$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.604 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	60.952 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	60.327 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	65.078 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	51.575 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	66.360 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	66.360 kgf/cm ²
App 27-4	= M.A.W.P.	46.886 kgf/cm ²

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 46.405 • 1.0
 = 60.327 kgf/cm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	1696.3	1829.3	0.927	60.62
Shell-1	1696.2	1829.3	0.927	60.63
Top Head	1696.3	1829.3	0.927	60.62

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	1407.13	1407.13	1.000
N1_24"	Nozzle	1407.13	1407.13	1.000
N4_1.5"	Nozzle	1407.13	1407.13	1.000
K1_2"	Nozzle	1407.13	1407.13	1.000
K2_2"	Nozzle	1407.13	1407.13	1.000
K3_2"	Nozzle	1407.13	1407.13	1.000
K4_2"	Nozzle	1407.13	1407.13	1.000
M1_24"	Nozzle	1407.13	1407.13	1.000
N2_24"	Nozzle	1407.13	1407.13	1.000
K5_2"	Nozzle	1407.13	1407.13	1.000
K6_2"	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom Head	1407.16	1407.16	1.000
Shell-1	1407.16	1407.16	1.000
Top Head	1407.16	1407.16	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N3_3"	206.70	1829.27	0.113
N1_24"	1295.70	1829.27	0.708
N4_1.5"	108.55	1829.27	0.059
K1_2"	128.24	1829.27	0.070
K2_2"	128.24	1829.27	0.070
K3_2"	128.24	1829.27	0.070
K4_2"	128.24	1829.27	0.070



M1_24"	1295.70	1829.27	0.708
N2_24"	1295.47	1829.27	0.708
K5_2"	129.72	1829.27	0.071
K6_2"	129.72	1829.27	0.071

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	60.952 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	60.327 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	65.291 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	51.575 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	66.360 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	66.360 kgf/cm ²
App 27-4	= M.A.W.P.	46.886 kgf/cm ²

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 46.405 • 1.0
 = 60.327 kgf/cm²

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	1803.6	1829.3	0.986	60.78
Shell-1	1801.8	1829.3	0.985	60.71
Top Head	1792.3	1829.3	0.980	60.40

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	1407.13	1407.13	1.000
N1_24"	Nozzle	1407.13	1407.13	1.000
N4_1.5"	Nozzle	1407.13	1407.13	1.000
K1_2"	Nozzle	1407.13	1407.13	1.000
K2_2"	Nozzle	1407.13	1407.13	1.000
K3_2"	Nozzle	1407.13	1407.13	1.000
K4_2"	Nozzle	1407.13	1407.13	1.000
M1_24"	Nozzle	1407.13	1407.13	1.000
N2_24"	Nozzle	1407.13	1407.13	1.000
K5_2"	Nozzle	1407.13	1407.13	1.000
K6_2"	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom Head	1407.16	1407.16	1.000
Shell-1	1407.16	1407.16	1.000
Top Head	1407.16	1407.16	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N3_3"	207.27	1829.27	0.113
N1_24"	1297.39	1829.27	0.709
N4_1.5"	108.69	1829.27	0.059
K1_2"	128.40	1829.27	0.070
K2_2"	128.40	1829.27	0.070
K3_2"	128.40	1829.27	0.070
K4_2"	128.40	1829.27	0.070



M1_24"	1297.39	1829.27	0.709
N2_24"	1290.93	1829.27	0.706
K5_2"	129.27	1829.27	0.071
K6_2"	129.27	1829.27	0.071

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	370	...	...	No Calc	No Calc
20	30	370	...	...	No Calc	No Calc
30	40	No Calc	3004	49	0.0022655	1085.65
40	50	360.833	3008	50	0.0023842	1096.44
50	60	No Calc	3004	49	0.0022655	1085.65

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	52	No Calc	...	19.6763
40	50	53	No Calc	...	24.3005
50	60	52	No Calc	...	19.6763

Minimum

20

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	370	No Calc	No Calc	No Calc
20	30	370	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	360.833	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	3784.34	...	3473.33	...	264.904
20	30	1007.93	...	892.733	...	70.5548
30	40	5092.33	3522781	4837.72	3544005	356.463
40	50	11534.8	20008346	10892.9	20091168	807.433
50	60	5092.33	3522781	4837.72	3544005	356.463
Total		26511	27053906.00	24934	27179180.00	1855

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Wght	399.993	140	30	...	Tailing+Bracing
30	Plat	4204.26	0.21131E-05	...	...	BTL PLT
30	Liqd	3521.19	...	-36.25	...	Liquid: 30
30	Insl	166.975	...	-33.75	...	Ins: 30
30	Nozl	74.066	...	-82.4006	...	N3_3"
30	Wght	9.99982	...	-40	...	Anti-Swirl Baff
40	Liqd	4627.94	...	35	...	Liquid: 40
40	Insl	371.194	...	151.25	...	Ins: 40
40	Nozl	4180.29	160.1	171.5	...	N1_24"
40	Nozl	15.2463	-160.1	135	...	N4_1.5"
40	Nozl	14.9948	160.1	103	...	K1_2"
40	Nozl	14.9948	160.1	10	...	K2_2"
40	Nozl	14.9948	-160.1	80	...	K3_2"
40	Nozl	14.9948	-160.1	10	...	K4_2"
40	Nozl	1170.61	...	95	160.1	M1_24"
40	Wght	99.9982	...	95	180	Manway Davit
40	Wght	144.997	...	277.5	...	Demister
40	Wght	49.9991	...	150	...	Pipe Support
40	Wght	49.9991	...	283	...	Demister Top Sp
40	Wght	49.9991	...	272	...	Demister bottom
40	Wght	399.993	150	260	...	Lifting Lug1
40	Wght	399.993	-150	260	...	Lifting Lug2
40	Wght	949.983	...	151.5	...	Cleat weight
50	Plat	4711.25	...	90	...	PTL-2
50	Insl	166.975	...	38.75	...	Ins: 50
50	Nozl	737.879	...	99.9	...	N2_24"
50	Nozl	10.7413	100	69.9	...	K5_2"
50	Nozl	10.7413	-100	69.9	...	K6_2"
50	Wght	14999.7	220	77.5	...	PSV Weight
50	Forc	...	...	65	...	N2 F&M

Total Weight of Each Detail Type:

Platforms	8915.5
Liquid	8149.1
Insulation	705.1
Nozzles	6259.6
Weights	17554.7
Sum of the Detail Weights	41584.0 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	28367.5	28367.5	28367.5	28367.5	28367.5	28367.5
Nozzles	6259.6	6259.6	6259.6	6259.6	6259.6	6259.6
Wld Weights	2410.0	2410.0	2410.0	2410.0	2410.0	2410.0



Platforms	...	...	8915.5	8915.5	8915.5	8915.5
Insulation	...	...	705.1	705.1	705.1	705.1
Empty Weights	...	...	...	145.0	145.0	...
Test Weights	...	145.0	...	...	...	...
Ope Weights	...	...	...	...	...	15144.7
Ope. Liquid	...	...	...	...	...	8149.1
Test Liquid	...	27041.7	...	...	...	...
Totals	37037.0	64223.7	46657.7	46802.7	46802.7	69951.5

Field Installation Options:**Miscellaneous Weight Percent: 7.0 %**

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	37037.0 kg
Shop Test Wt.	- Fabricated weight + water (full)	64223.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	46657.7 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	46802.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	46802.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	69951.5 kg
Field Test Wt.	- Empty weight + test medium (full)	73844.4 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	42200.8 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	37192.3 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	27978.7 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	252009	249634
20	30	93336.7	92457.1
30	40	103608	95718.5
40	50	285860	275786
50	60	103608	95718.5

Total	838420.9	809314.5 cm ²
Total	902.5	871.1 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	4449.24	4449.24	4449.24	559.991	4449.24
20	30	1078.48	1078.48	1078.48	...	1078.48
30	40	9904.1	13425.3	13425.3	0.88839E-04	9904.1
40	50	20284.5	24912.4	40283.8	6668.25	20284.5
50	60	11086.4	26086.1	14607.6	32999.5	26086.1

Empty Support Force + the Sum of the Y forces	51848.51 kgf
Operating Support Force + the Sum of the Y forces	74997.38 kgf
Hydro Support Force + the Sum of the Y forces	78890.21 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid	Cumulative Oper. Wgt.	Cumulative Hydro. Wgt.
------	----	----------------------------------	--------------------------	---------------------------



		kg	kg	kg
10	20	61802.4	69951.5	73844.4
20	30	57353.2	65502.3	69395.1
30	40	56274.7	64423.8	68316.7
40	50	46370.6	50998.5	54891.4
50	60	26086.1	26086.1	14607.6

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	40227.7	40227.7	40227.7
20	30	39667.7	39667.7	39667.7
30	40	39667.7	39667.7	39667.7
40	50	39667.7	39667.7	39667.7
50	60	32999.5	32999.5	32999.5

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3_3"	95.25	104.01	100	600	GR 1.1	...	...	...	...
N1_24"	47.62	52.01	100	300	GR 1.1	14.387	47.623	33.236	47.623
N4_1.5"	142.87	156.12	100	900	GR 1.1	...	...	...	...
K1_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
K2_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
K3_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
K4_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
M1_24"	47.62	52.01	100	300	GR 1.1	...	...	...	...
N2_24"	47.62	52.01	100	300	GR 1.1	...	...	...	...
K5_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
K6_2"	47.62	52.01	100	300	GR 1.1	...	...	...	...
Min Rating	47.623	52.007, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N1_24"	657.225	0.50	7266	4587	46.886	47.623	Pass

Evaluating Flange on Nozzle [N124"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR] \\
 &= 16(7266) + 4 \cdot \text{abs}(4587) \cdot 657.225 \leq \pi \cdot 657.225^3 \cdot [(47.623 - 46.886) + 500 \cdot 47.623] \\
 &= 0.12831E+06 \leq 0.21893E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N124" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(47.623, 0.500 \cdot 47.623 + 47.623 - 128310.1 / (\pi \cdot 657.225^3)) \\
 &= \min(47.623, 57.047) \\
 &= 47.623 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 11.1485 Hz.

The Natural Frequency for the Vessel (Ope...) is 10.672 Hz.

The Natural Frequency for the Vessel (Filled) is 10.9863 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	4757.29	11084.5

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	4757.29	11084.5

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
50	65.00	3364.	-5046.	3364.	5651.		5651.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.943) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.895)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	10.672 Hz
Natural Frequency of Vessel (Empty)	11.148 Hz
Natural Frequency of Vessel (Test)	10.986 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.527
Structure Height to Diameter ratio	2.599
Height to top of Structure	764.200 cm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 27432.0$ cm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 135.0 + 30.0 = 165.0$ cm
 $= 5.413$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
Because z (5.413 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 27432.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.527 \cdot 112.248$$

$$= 804.0 \text{ lbs. } [364.7 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Skirt 2	165.0	0.000	0.000	0.000	0.849	1.000	78.119
Skirt 1	350.0	0.000	0.000	0.000	0.849	1.000	78.119
Bottom Head	402.5	0.000	0.000	0.000	0.849	1.000	78.119
Shell-1	556.3	0.000	0.000	0.000	0.885	1.000	78.119
Top Head	744.5	0.000	0.000	0.000	0.941	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/682.5 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[9.92/22.39^2] \cdot 10^4 = 197.909$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.0004 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 68264 / (682.5 \cdot 305.361^2)$$

$$= 0.10727E-02$$

Since Vp is > 0.0004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area	Elevation	Pressure	Force	Cf
----	-----------	-----------	----------	-------	----



	cm ²	cm	kgf/m ²	kgf	
BTL PLT	46256.00	400.00	78.12	361.35	1.00
PTL-2	66256.01	797.50	78.12	517.58	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Tailing+Braci	0.00	60.00	78.12	0.00
Anti-Swirl Ba	0.00	360.00	78.12	0.00
Manway Davit	0.00	500.00	78.12	0.00
Demister	0.00	682.50	78.12	0.00
Pipe Support	0.00	555.00	78.12	0.00
Demister Top	0.00	688.00	78.12	0.00
Demister bott	0.00	677.00	78.12	0.00
Lifting Lug1	0.00	665.00	78.12	0.00
Lifting Lug2	0.00	665.00	78.12	0.00
Cleat weight	0.00	556.50	78.12	0.00
PSV Weight	0.00	785.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 10.672 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	165	386.23	104282	78.1189	401.148
20	30	350	386.23	38623	78.1189	148.573
30	40	402.5	400.92	2004.6	78.1189	402.723
40	50	556.25	401.44	121436	78.1189	467.133
50	60	744.487	400.92	27730.4	78.1189	716.556

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(24.77030.75)$
= 0.222 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/10.672)
= 0.094

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.222$, 0.094) per 12.8.2
= 0.094

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 2.0)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.094(2.0 / 2.0))$
= 1.110

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 2.0), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1568 \cdot 69951.5$
= 10968.402 kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 69952$



= 2193.7 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 10.672 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	135	4449.24	179.048	192.921	-2852.16
20	30	320	1078.48	102.875	110.846	-2991.69
30	40	372.5	13425.3	1490.73	1184.95	-3025.51
40	50	526.25	24912.4	3908.03	3428.59	-3446.53
50	60	680	26086.1	5287.72	2421.35	-4227.78



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	135	4757.29	4757.29	43315.1	43315.1
20	30	320	4757.29	4757.29	30470.4	30470.4
30	40	372.5	4757.29	4757.29	25713.1	25713.1
40	50	526.25	4757.29	4757.29	25475.3	25475.3
50	60	680	4757.29	4757.29	11084.5	11084.5

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	135	2136.13	10968.4	10095.1	64470
20	30	320	1734.98	10789.4	4869.11	35097
30	40	372.5	1586.41	10686.5	3208.41	24359.1
40	50	526.25	1183.69	9195.75	3139.16	23862
50	60	680	716.556	5287.72	265.034	1955.78

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	4757.29	0.013607	0.051599	0.36125E-04
20	30	4757.29	0.071007	0.092963	0.46313E-04
30	40	4757.29	0.094121	0.095281	0.46439E-04
40	50	4757.29	0.16804	0.24481	0.51726E-04
50	60	4757.29	0.2461	0.24739	0.51726E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 34.125 Actual deflection : 0.247 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	2136.13	0.0030809	0.011121	0.73694E-05
20	30	1734.98	0.015017	0.019285	0.88562E-05
30	40	1586.41	0.019507	0.019728	0.88720E-05
40	50	1183.69	0.033434	0.047485	0.93386E-05
50	60	716.556	0.047719	0.047952	0.93386E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	739.936	4624.6
20	30	739.936	4624.6
30	40	768.079	4800.49
40	50	769.075	4806.72
50	60	768.079	4800.49

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 34.125 Actual deflection : 0.048 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 34.125 Actual deflection : 0.295 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	1300.56	1151.92	95692632	84801272
20	30	1300.56	1151.92	95692632	84801272
30	40	4822.47	4548.87	349845920	330656992
40	50	4920.2	4646.41	357190592	337987424
50	60	4822.47	4548.87	349845920	330656992

Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1182.01	1536.62	-1059.8	-1059.8
20	30	844.296	1097.58	-1059.8	-1059.8
30	40	1688.59	2195.17	-1428.9	-1428.9
40	50	1688.59	2195.17	-1432.11	-1432.11
50	60	1688.59	2195.17	-1428.9	-1428.9

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	678.749	...	882.373
40	50	665.452	...	865.088
50	60	678.749	...	882.373

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-53.6518	-60.7262	-64.1057	47.4376	47.4376
20	30	-49.7893	-56.8638	-60.2432	46.7773	46.7773
30	40	-12.3712	-12.3712	-12.3712	11.9966	11.9966
40	50	-9.97988	-9.97988	-9.97988	11.7364	11.7364
50	60	-5.73465	-5.73465	-5.73465	9.97997	9.97997

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	47.4376	11.9045	76.0248	5.95223	...
20	30	46.7773	5.74179	41.3874	2.87089	...
30	40	11.9966	0.97031	7.36688	0.48516	...
40	50	11.7364	0.92878	7.06004	0.46439	...
50	60	9.97997	0.080154	0.59148	0.040077	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	47.6082
20	30	...	...	...	24.5496
30	40	...	...	...	4.15175
40	50	...	...	...	3.9664
50	60	...	...	...	0.27085

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	-4.38039	-2.47601
20	30	-4.38039	-2.59714
30	40	-1.10925	-0.66511

40	50	-1.08597	-0.74176
50	60	-1.10925	-0.92941

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	51.0784	51.0784	45.2649	45.2649
20	30	35.9316	35.9316	31.842	31.842
30	40	7.77638	7.77638	7.34985	7.34985
40	50	7.53735	7.53735	7.13213	7.13213
50	60	3.35226	3.35226	3.16839	3.16839

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.39	1182.01	-168.45	1059.80	0.0443	0.1589
20	34.28	844.30	-142.62	1059.80	0.0406	0.1346
30	7.26	1688.59	-34.22	1428.90	0.0043	0.0240
40	9.14	1688.59	-31.27	1432.11	0.0054	0.0218
50	6.57	1688.59	-20.26	1428.90	0.0039	0.0142

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	90.00	1182.01	-202.25	1059.80	0.0761	0.1908
20	54.87	844.30	-159.64	1059.80	0.0650	0.1506
30	10.89	1688.59	-36.96	1428.90	0.0064	0.0259
40	12.52	1688.59	-33.96	1432.11	0.0074	0.0237
50	6.94	1688.59	-20.27	1428.90	0.0041	0.0142

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	45.31	1182.01	-175.53	1059.80	0.0383	0.1656
20	27.21	844.30	-149.69	1059.80	0.0322	0.1412
30	7.26	1688.59	-34.22	1428.90	0.0043	0.0240
40	9.14	1688.59	-31.27	1432.11	0.0054	0.0218



50	6.57	1688.59	-20.26	1428.90	0.0039	0.0142
----	------	---------	--------	---------	--------	--------

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.34	1182.01	-237.74	1059.80	0.0942	0.2243
20	64.64	844.30	-183.56	1059.80	0.0766	0.1732
30	14.10	1688.59	-40.18	1428.90	0.0084	0.0281
40	15.61	1688.59	-37.06	1432.11	0.0092	0.0259
50	7.26	1688.59	-20.59	1428.90	0.0043	0.0144

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1536.62	-117.50	1059.80		0.1109
20		1097.58	-109.89	1059.80		0.1037
30	0.11	2195.17	-24.85	1428.90	0.0001	0.0174
40	2.22	2195.17	-22.18	1432.11	0.0010	0.0155
50	4.29	2195.17	-15.75	1428.90	0.0020	0.0110

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1536.62	-111.54	1059.80		0.1052
20		1097.58	-107.02	1059.80		0.1010
30		2195.17	-24.37	1428.90		0.0171
40	1.76	2195.17	-21.72	1432.11	0.0008	0.0152
50	4.25	2195.17	-15.71	1428.90	0.0019	0.0110

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	45.31	1182.01	-175.53	1059.80	0.0383	0.1656
20	27.21	844.30	-149.69	1059.80	0.0322	0.1412
30	686.01	1688.59		1428.90	0.4063	
40	674.59	1688.59		1432.11	0.3995	
50	685.32	1688.59		1428.90	0.4059	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.34	1182.01	-237.74	1059.80	0.0942	0.2243
20	64.64	844.30	-183.56	1059.80	0.0766	0.1732
30	692.85	1688.59		1428.90	0.4103	
40	681.06	1688.59		1432.11	0.4033	
50	686.01	1688.59		1428.90	0.4063	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	45.31	1182.01	-175.53	1059.80	0.0383	0.1656
20	27.21	844.30	-149.69	1059.80	0.0322	0.1412
30	7.26	1688.59	-34.22	1428.90	0.0043	0.0240
40	9.14	1688.59	-31.27	1432.11	0.0054	0.0218
50	6.57	1688.59	-20.26	1428.90	0.0039	0.0142

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.34	1182.01	-237.74	1059.80	0.0942	0.2243
20	64.64	844.30	-183.56	1059.80	0.0766	0.1732
30	14.10	1688.59	-40.18	1428.90	0.0084	0.0281
40	15.61	1688.59	-37.06	1432.11	0.0092	0.0259



50	7.26	1688.59	-20.59	1428.90	0.0043	0.0144
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Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1536.62	-117.50	1059.80		0.1109
20		1097.58	-109.89	1059.80		0.1037
30	882.48	2195.17		1428.90	0.4020	
40	867.31	2195.17		1432.11	0.3951	
50	886.66	2195.17		1428.90	0.4039	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1536.62	-111.54	1059.80		0.1052
20		1097.58	-107.02	1059.80		0.1010
30	882.00	2195.17		1428.90	0.4018	
40	866.84	2195.17		1432.11	0.3949	
50	886.62	2195.17		1428.90	0.4039	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-101.09	1059.80		0.0954
20		844.30	-96.57	1059.80		0.0911
30	678.37	1688.59		1428.90	0.4017	
40	667.21	1688.59		1432.11	0.3951	
50	682.99	1688.59		1428.90	0.4045	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-53.79	1059.80		0.0508
20		844.30	-50.36	1059.80		0.0475
30	667.08	1688.59		1428.90	0.3951	
40	656.03	1688.59		1432.11	0.3885	
50	673.34	1688.59		1428.90	0.3988	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-108.16	1059.80		0.1021
20		844.30	-103.64	1059.80		0.0978
30	678.37	1688.59		1428.90	0.4017	
40	667.21	1688.59		1432.11	0.3951	
50	682.99	1688.59		1428.90	0.4045	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-101.09	1059.80		0.0954
20		844.30	-96.57	1059.80		0.0911
30	678.37	1688.59		1428.90	0.4017	
40	667.21	1688.59		1432.11	0.3951	
50	682.99	1688.59		1428.90	0.4045	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-108.16	1059.80		0.1021
20		844.30	-103.64	1059.80		0.0978
30		1688.59	-24.37	1428.90		0.0171
40	1.76	1688.59	-21.72	1432.11	0.0010	0.0152



50 4.25 1688.59 -15.71 1428.90 0.0025 0.0110

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.31	985.01	-161.72	883.17	0.0359	0.1831
20	23.25	703.58	-142.17	883.17	0.0330	0.1610
30	685.49	1407.16		1190.75	0.4871	
40	674.00	1407.16		1193.43	0.4790	
50	685.42	1407.16		1190.75	0.4871	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.31	985.01	-161.72	883.17	0.0359	0.1831
20	23.25	703.58	-142.17	883.17	0.0330	0.1610
30	6.74	1407.16	-32.81	1190.75	0.0048	0.0276
40	8.55	1407.16	-30.00	1193.43	0.0061	0.0251
50	6.67	1407.16	-20.00	1190.75	0.0047	0.0168

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1182.01	-108.16	1059.80		0.1021
20		844.30	-103.64	1059.80		0.0978
30	678.37	1688.59		1428.90	0.4017	
40	667.21	1688.59		1432.11	0.3951	
50	682.99	1688.59		1428.90	0.4045	

Absolute Maximum of the all of the Stress Ratio's 0.4871

Governing Element: Bottom Head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt 2	...	-63.2	47.4	51.1	35.3	+ Tens
Skirt 2	...	-63.2	-47.4	-51.1	-161.7	- Comp
Skirt 1	...	-59.5	46.8	35.9	23.2	+ Tens
Skirt 1	...	-59.5	-46.8	-35.9	-142.2	- Comp
Bottom Head	678.7	-13.0	12.0	7.8	685.5	+ Tens
Bottom Head	678.7	-13.0	-12.0	-7.8	...	- Comp
Shell-1	665.5	-10.7	11.7	7.5	674.0	+ Tens
Shell-1	665.5	-10.7	-11.7	-7.5	...	- Comp
Top Head	678.7	-6.7	10.0	3.4	685.4	+ Tens
Top Head	678.7	-6.7	-10.0	-3.4	...	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

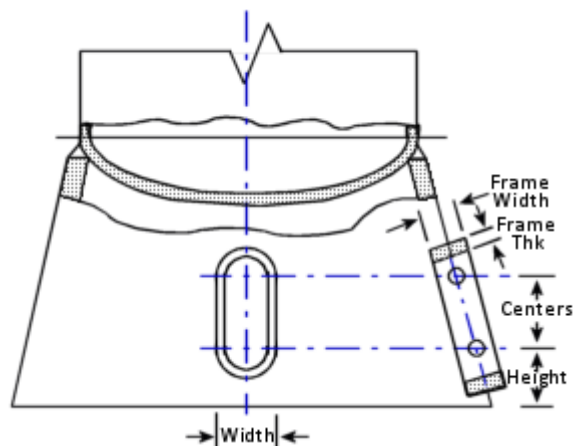
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt 2

Total vessel (empty) weight	Fg	57010.145 kgf
Weight of vessel contents (liquid)	Ff	8149.136 kgf
Vessel Gross Weight	Fg + Ff	65159.277 kgf
Skirt Mean Diameter	Dz	2957.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	2700.000 mm
Additional Axial Force at section 1-1	F1	-5045.840 kgf
Moment at section 1-1	M1	65567.469 kgf-m.
Applied Moment - Bottom of Skirt	Mb	107785.164 kgf-m.
Weight at Bottom of Skirt	W	69951.539 kgf

**Skirt Access Opening Input Data:**

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	200.0000	0.0	2050.00	0.00	12.000	12.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-3.8782	3.8782

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-356.122	-3.878



Height of Weakest Section	H	2049.865	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-28.689	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	12589703.000	cm ⁴
Principle Moment of Inertia	Iyy	12101940.000	cm ⁴
Distance to Extreme Fiber	ybar	1484.700	mm
Distance to Extreme Fiber	xbar	1513.389	mm
Minimum Section Modulus	W4	79965872.000	mm ³
Metal Area at the Cutting Plane	A4	1129.983	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / \text{Dz})$$

$$= \sin^{-1}(200.0/2957.0)$$

$$= 0.0677 \text{ rad (3.8782 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= \text{Dz} / 2$$

$$= 2957.0/2$$

$$= 1478.5 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.068 \cdot \sqrt{(1478.5/12.4)}$$

$$= 0.739$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= \text{ex}(\text{new}) - \text{ci} - \text{cext}$$

$$= 14.0 - 0.8 - 0.8$$

$$= 12.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (\text{Mbottom} - \text{Mtop}) \cdot (\text{H} - \text{CutHeight}) / \text{H} + \text{Mtop}$$

$$= (107785.164 - 65567.469) \cdot (2700.0 - 2049.865) / 2700.0 + 65567.469$$

$$= 75733.102 \text{ kgf-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$\text{Zxx} = \text{Ixx} / \text{ybar} = 12589703 / 1484.7 = 84796352 \text{ mm}^3$$

$$\text{Zyy} = \text{Iyy} / \text{xbar} = 12101940 / 1513.389 = 79965872 \text{ mm}^3$$

$$\text{Z} = \min(\text{Zxx}, \text{Zyy}) = \min(84796352, 79965872) = 79965872 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(\text{Ixx}, \text{Iyy}) / \text{Area}(\text{A4}))}$$

$$= \sqrt{(\min(12589703, 12101940) / 1129.983)}$$

$$= 1034.884 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= \text{rg} / \text{L}$$

$$= 1034.884 / 2700.0$$

$$= 0.383$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(\text{ex}^2 + \text{ey}^2)}$$

$$= \sqrt{(-28.689^2 + 0.0^2)}$$

$$= 0.000 \text{ mm}$$

$$\text{Where } \text{ex} = -28.689 : \text{ey} = 0.0$$

Moment Increment at Centroid [ΔM4]:



$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-5045.84 - 57010.14 - 8149.14) \cdot 28.689 \\
 &= 2014.141 \text{ kgf-m.}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(12.0 - 12.4) / 2 \\
 &= 0.200 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 200.0 - 12.0 \\
 &= 188.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(188.0 / 2957.0) \\
 &= 0.064 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 12.0 \cdot 12.0 \\
 &= 2.880 \text{ cm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.064) \cdot 2957.0 \cdot 12.4 \\
 &= 1128.594 \text{ cm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 1128.594 + 2.88 \\
 &= 1131.474 \text{ cm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 2957.0 \cdot \cos(0.064) + 0.2 - 0.5 \cdot 12.0 \\
 &= 1469.709 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot et \cdot yt) / A4 \\
 &= (0.5 \cdot 2957.0 \cdot 12.4 \cdot 188.0 - 2 \cdot 12.0 \cdot 12.0 \cdot 1469.709) / 1131.474 \\
 &= 26.721 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 2957.0 \cdot \cos(0.064) + 0.2 + 26.721); \text{abs}(0.5 \cdot 2957.0 - 26.721)] \\
 &= 1502.430 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.064 - \sin(0.064) \cdot \cos(0.064)) \cdot 12.4 \cdot (2957.0/2)^3 \\
 &= 12081028.0 \text{ cm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (12081028 + 2.88(1469.709^2 + 12.0^2/12) - 1131.474 \cdot 26.721^2) / 1502.43 \\
 &= 80770296.000 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 1131.474 / (\pi \cdot 2957.0 \cdot 12.4)) \\
 &= 0.982
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 80770296 / (\pi \cdot 2957.0^2 \cdot 12.4)) \\
 &= 0.949
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 2957.0^2 \cdot 12.4 \cdot 1059.8 \\ &= 902483.6 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 2957.0 \cdot 12.4 \cdot 1059.8 \\ &= 1220808.2 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max}) \\ &= 70205.1 / (0.982 \cdot 1220808) + (\text{abs}(75733.1) + 70205.1 \cdot \text{abs}(26.721)) / \\ &\quad (0.949 \cdot 902484) \\ &= 0.149 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt 1

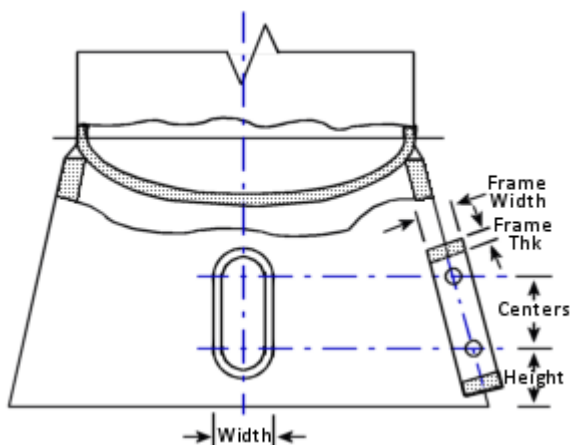
Total vessel (empty) weight	Fg	57010.145 kgf
Head Weight below section 2-2	DeltaFg	9904.098 kgf
Weight of vessel contents (liquid)	Ff	8149.136 kgf
Vessel Gross Weight	Fg + Ff	65159.277 kgf
Skirt Mean Diameter	Dz	2957.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-5045.840 kgf
Moment at section 1-1	M1	50072.238 kgf-m.
Applied Moment - Bottom of Skirt	Mb	65567.469 kgf-m.
Weight at Bottom of Skirt	W	65502.309 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 2955.000 mm
Head Corroded Thickness	eb 49.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 728.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 2906.0 / (2 * 728.0) = 1.9959 \\
 R &= D_i (0.44 * K + 0.02) = 2906.0 * (0.44 * 1.996 + 0.02) = 2610.1309 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 2906.0 * ((0.5 / 1.996) - 0.08) = 495.5201 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	80.0000	0.0	500.00	0.00	12.000	12.000
2	80.0000	60.0	500.00	0.00	12.000	12.000
3	80.0000	120.0	500.00	0.00	12.000	12.000
4	80.0000	180.0	500.00	0.00	12.000	12.000
5	80.0000	240.0	500.00	0.00	12.000	12.000
6	80.0000	300.0	500.00	0.00	12.000	12.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	1548.2556	1468.2557
2	3	1548.2556	1468.2557
3	4	1548.2556	1468.2557
4	5	1548.2556	1468.2557
5	6	1548.2556	1468.2557
1	6	1548.2556	1468.2557

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1548.2556 mm
 Minimum Ligament between Openings is: 1468.2557 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(2957.0 \cdot 12.4)} \text{ mm}$$

$$= 382.972 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 6

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-1.5503	1.5503
2	58.4497	61.5503
3	118.4497	121.5503
4	178.4497	181.5503
5	238.4497	241.5503
6	298.4497	301.5503

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
1.550	58.450
61.550	118.450
121.550	178.450
181.550	238.450
241.550	298.450
-58.450	-1.550

Height of Weakest Section	H	499.988	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-0.000	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	12128485.000	cm ⁴
Principle Moment of Inertia	Iyy	12128485.000	cm ⁴
Distance to Extreme Fiber	ybar	1484.700	mm
Distance to Extreme Fiber	xbar	1484.700	mm
Minimum Section Modulus	W4	81689872.000	mm ³
Metal Area at the Cutting Plane	A4	1109.675	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(80.0/2957.0)$$

$$= 0.0271 \text{ rad (1.5503 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 2957.0/2$$

$$= 1478.5 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2.9	8.4	...	...
2.9	8.4	...	...
2.9	8.4	...	...
2.9	8.4	...	...
2.9	8.4	...	...
2.9	8.4	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / ez)}$$

$$= 0.027 \cdot \sqrt{(1478.5/12.4)}$$

$$= 0.295$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(new) - ci - cext$$

$$= 14.0 - 0.8 - 0.8$$

$$= 12.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (65567.469 - 50072.238) \cdot (1000.0 - 499.988) / 1000.0 + 50072.238$$

$$= 57820.039 \text{ kgf-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 12128485 / 1484.7 = 81689872 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 12128485 / 1484.7 = 81689872 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(81689872, 81689872) = 81689872 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A4))}$$

$$= \sqrt{(\min(12128485, 12128485) / 1109.675)}$$

$$= 1045.456 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1045.456 / 1000.0$$

$$= 1.045$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(ex^2 + ey^2)}$$

$$= \sqrt{(-0.0^2 + -0.0^2)}$$

$$= 28.689 \text{ mm}$$

$$\text{Where } ex = -0.0 : ey = -0.0$$

Moment Increment at Centroid [ΔM4]:

$$= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$$

$$= \text{abs}(-5045.84 - 57010.14 - 8149.14) \cdot 0.0$$

$$= 0.000 \text{ kgf-m.}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:



$$\begin{aligned}
 &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\
 &= -5045.84 - 57010.145 - 8149.136 + 4 \cdot 50072.238 / 2957.0 \\
 &= -2471.4 \text{ kgf}
 \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned}
 &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\
 &= -5045.84 - 57010.145 - 8149.136 - 4 \cdot 50072.238 / 2957.0 \\
 &= -137938.9 \text{ kgf}
 \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned}
 &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\
 &= (-2471.361 + 9904.098 + 8149.136) / (\pi \cdot 2955.0 \cdot 49.0) + 46.405 \cdot 2955.0 / (4 \cdot 49.0) \\
 &= 703.06 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned}
 &= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\
 &= (-0.1E+06 + 9904.098 + 8149.136) / (\pi \cdot 2955.0 \cdot 49.0) + 46.405 \cdot 2955.0 / (4 \cdot 49.0) \\
 &= 673.28 \text{ kgf/cm}^2
 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned}
 &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\
 &= (8149.136 + 9904.098) / (\pi \cdot 2955.0 \cdot 49.0) + 46.405 \cdot 2955.0 / (4 \cdot 49.0) \\
 &= 703.60 \text{ kgf/cm}^2
 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned}
 &= Fzp / (\pi \cdot Dz \cdot ez) \\
 &= -2471.361 / (\pi \cdot 2957.0 \cdot 12.4) \\
 &= -2.15 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned}
 &= Fzq / (\pi \cdot Dz \cdot ez) \\
 &= -137938.891 / (\pi \cdot 2957.0 \cdot 12.4) \\
 &= -119.75 \text{ kgf/cm}^2
 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$\begin{aligned}
 &= 1 - 0.5 (Db + eb - Dz + ez) / (r + eb) \\
 &= 1 - 0.5 (2955.0 + 49.0 - 2957.0 + 12.4) / (495.52 + 49.0) \\
 &= 0.9455
 \end{aligned}$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.945) = 19.011 \text{ deg}$$

Dimension [a]:

$$\begin{aligned}
 &= 0.5 \cdot \sqrt{eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)} \\
 &= 0.5 \cdot \sqrt{49.0^2 + 12.4^2 + 2 \cdot 49.0 \cdot 12.4 \cdot 0.945} \\
 &= 30.429 \text{ mm}
 \end{aligned}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned}
 Mp &= a \cdot Fzp = 30.42 \cdot -2471.4 = -75.20 \text{ kgf-m.} \\
 Mq &= a \cdot Fzq = 30.42 \cdot -137938.9 = -4197.34 \text{ kgf-m.}
 \end{aligned}$$

Correction factor [C]:

$$\begin{aligned}
 &= 0.63 - 0.057 (eb/ez)^2 \\
 &= 0.63 - 0.057 (49.0/12.4)^2 \\
 &= -0.260
 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned}
 &= C \cdot 6 \cdot Mp / (\pi \cdot Db \cdot eb^2) \\
 &= -0.26 \cdot 6 \cdot -75.201 / (\pi \cdot 2955.0 \cdot 49.0^2)
 \end{aligned}$$



$$= 0.53 \text{ kgf/cm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e b^2)$$

$$= -0.26 \cdot 6 \cdot -4197.337 / (\pi \cdot 2955.0 \cdot 49.0^2)$$

$$= 29.38 \text{ kgf/cm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e z^2)$$

$$= -0.26 \cdot 6 \cdot -75.201 / (\pi \cdot 2957.0 \cdot 12.4^2)$$

$$= 8.22 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e z^2)$$

$$= -0.26 \cdot 6 \cdot -4197.337 / (\pi \cdot 2957.0 \cdot 12.4^2)$$

$$= 458.53 \text{ kgf/cm}^2$$

Geometry variables, 16.12-32 to 36:

$$e b / D_b = 1.929 / 116.339 = 0.0166$$

$$y = 125 \cdot e b / D_b = 125 \cdot 49.0 / 2955.0 = 2.0728$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 2.073 = 3.7854$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + P_h) \cdot D_b ((y / y_A) \cdot \alpha - 1) / (4 \cdot e b)$$

$$= (46.405 + 0.0) \cdot 2955.0 ((19.01 / 40.0) \cdot 3.785 - 1) / (4 \cdot 49.0)$$

$$= 559.09 \text{ kgf/cm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 703.057 - 0.526 + 559.088$$

$$= 1261.62 \text{ kgf/cm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 703.057 + 0.526 - 559.088$$

$$= 144.49 \text{ kgf/cm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 673.276 - 29.384 + 559.088$$

$$= 1202.98 \text{ kgf/cm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 673.276 + 29.384 - 559.088$$

$$= 143.57 \text{ kgf/cm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 703.6 + 0.526 + 559.088$$

$$= 1263.21 \text{ kgf/cm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 703.6 - 0.526 - 559.088$$

$$= 143.99 \text{ kgf/cm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 703.6 + 29.384 + 559.088$$

$$= 1292.07 \text{ kgf/cm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 703.6 - 29.384 - 559.088$$

$$= 115.13 \text{ kgf/cm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -2.145 - 8.215$$

$$= -10.36 \text{ kgf/cm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -2.145 + 8.215$$

$$= 6.07 \text{ kgf/cm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -119.747 - 458.531$$

$$= -578.28 \text{ kgf/cm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -119.747 + 458.531$$

$$= 338.78 \text{ kgf/cm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (703.057 / 1407.16)^2$$

$$= 3987.30 \text{ kgf/cm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (673.276 / 1407.16)^2$$

$$= 4006.72 \text{ kgf/cm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (703.6 / 1407.16)^2$$

$$= 3986.94 \text{ kgf/cm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (703.6 / 1407.16)^2$$

$$= 3986.94 \text{ kgf/cm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (-2.145 / 1407.16)^2$$

$$= 4221.48 \text{ kgf/cm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2$$

$$= 1407.16(3 - (1 / 1.5)) \cdot (-119.747 / 1407.16)^2$$

$$= 4214.69 \text{ kgf/cm}^2$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$$

$$= \min(1; 1109.675 / (\pi \cdot 2957.0 \cdot 12.4))$$

$$= 0.963$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3)))$$

$$= \min(1; 4 \cdot 0 \cdot 81689872 / (\pi \cdot 2957.0^2 \cdot 12.4))$$

$$= 0.959$$

Maximum Bending Moment [Mmax]:



$$\begin{aligned}
 &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\
 &= \pi/4 \cdot 2957.0^2 \cdot 12.4 \cdot 1059.8 \\
 &= 902483.6 \text{ kgf-m.}
 \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned}
 &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\
 &= \pi \cdot 2957.0 \cdot 12.4 \cdot 1059.8 \\
 &= 1220808.2 \text{ kgf}
 \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned}
 &= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max}) \\
 &= 70205.1 / (0.963 \cdot 1220808) + (\text{abs}(57820.0) + 70205.1 \cdot \text{abs}(0.0)) / \\
 &\quad (0.959 \cdot 902484) \\
 &= 0.126 \text{ must be } \leq \text{ to } 1, \text{ Passed}
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	1261.62	3987.30	Pass	46
Stot1po	144.49	3987.30	Pass	47
Stot1qi	1202.98	4006.72	Pass	48
Stot1qo	143.57	4006.72	Pass	49
Stot2pi	1263.21	3986.94	Pass	50
Stot2po	143.99	3986.94	Pass	51
Stot2qi	1292.07	3986.94	Pass	52
Stot2qo	115.13	3986.94	Pass	53
Stot3pi	-10.36	4221.48	Pass	54
Stot3po	6.07	4221.48	Pass	55
Stot3qi	-578.28	4214.69	Pass	56
Stot3qo	338.78	4214.69	Pass	57

The stresses are satisfactory in the region without openings.

The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	580.052 cm
Center of Gravity of the Liquid	377.053 cm
Center of Gravity of the Insulation	526.250 cm
Center of Gravity of the Nozzles	555.792 cm
Center of Gravity of the Added Weights (Operating)	716.695 cm
Center of Gravity of the Added Weights (Empty)	491.815 cm
Center of Gravity of Bare Shell New and Cold	462.561 cm
Center of Gravity of Bare Shell Corroded	464.365 cm
Vessel CG in the Operating Condition	542.963 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	499.968 cm
Vessel CG in the Test Condition	509.592 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2971.0002	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1407.13	kgf/cm ²
Basing Yield Stress		2671.52	kgf/cm ²
Inside Diameter of Basing	DI, D4	2791.0000	mm
Outside Diameter of Basing	DOU, D5	3251.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	1407.16	kgf/cm ²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	3111.0002	mm
Bolt Allowable Shear Stress		844.296	kgf/cm ²
Ultimate Comp. Strength of Concrete	FPC	418.1	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	130.0000	mm
Width of Gussets at Base Plate	BWDT	130.0000	mm
Gusset Plate Elastic Modulus	E	2057170.1	kgf/cm ²
Gusset Plate Yield Stress	SY	2671.5	kgf/cm ²
Height of Gussets	HG, h1	230.0000	mm
Distance between Gussets	RG, b6	90.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	46802.7	kgf
Operating Weight of Vessel	ROW	74997.4	kgf
Earthquake Moment on Basing	EQMOM	98029.5	kgf-m.
Wind Moment on Basing	WIMOM	43654.6	kgf-m.
User Moment at the Base [Earthquake Case]		43315.1	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		43315.1	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 141345/3111.0) - 72804) / 16$$

$$= 6808.2 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 86970/3111.0) - 46803) / 16$$

$$= 4063.7 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3111.0) - 74997) / 16$$

$$= -4687.3 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	8.5561	cm ²
Area Available in all the Bolts	Abss * RN :	136.8979	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.451	1.511	2.441	0.438	0.781	17381.7	90185.4	2108.4
4	0.376	0.338	1.705	2.277	0.421	0.784	18855.8	91659.5	2451.6
6	0.319	0.310	1.561	2.400	0.433	0.782	17744.7	90548.4	2189.1
8	0.315	0.312	1.548	2.410	0.434	0.782	17653.6	90457.3	2168.6
10	0.313	0.313	1.545	2.413	0.435	0.782	17630.9	90434.6	2163.5
12	0.313	0.313	1.544	2.414	0.435	0.782	17625.2	90428.9	2162.3

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 17625.2 / (1.401 \cdot 3111.0 \cdot 2.414)$$

$$= 335.153 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.2 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 8.556 / (\pi \cdot 3111.0)$$

$$= 1.401 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 15.532[(2 \cdot 0.3131 \cdot 3111.0 + 230.0) / (2 \cdot 0.3131 \cdot 3111.0)]$$

$$= 17.366 \text{ kgf/cm}^2, \text{ Should be } \leq 229.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 140.0, b = 305.422, l/b = 0.458382

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(387.676, 1178.434) = 1178.434 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1178.434 / 1763.2)} + 1.6$$

$$= 21.625 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 90.0, b/l = 0.642857

As b/l (0.643) is less than 1, inverting b/l = 1.556.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 12039.81 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0/140.0)) / (\pi \cdot 35.0))) + 1] - [(0.7 - 0.065) \cdot 12039.81 / (4 \cdot \pi)]$$

$$= 1513.4572 \text{ kgf}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(1513.46) / 1781.01) + 1.6}$$

$$= 24.1802 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 12039.81 / (1265.526 \cdot 130.0) + 1.6$$

$$= 11.125 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3111.0 / 16 = 610.843 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba)) \cdot 1.5 \cdot \text{Sktope})^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (12040 \cdot 70.0 / (610.84 \cdot 262.0 \cdot 2111))^{2/3} \cdot 1485.51^{1/3} + \text{Ca}$$

$$= 7.959 + 1.6 = 9.559 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 15725.7 / (8.556 \cdot 16)$$

$$= 114.9 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	21.6252	mm
Actual Basing Thickness as entered by user	32.0000	mm
Required Thickness of Chair Cap	24.1802	mm
Actual Top Ring Thickness as entered by user	38.0000	mm
Required Gusset thickness, + CA	11.1250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	9.5586	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	102.7183	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 237.743 \cdot (14.0 - 1.6)$$

$$= 29.48 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 29 / [(0.4 \cdot 2437) \cdot 2 \cdot 0.707]$$

$$= 2.139 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 12039.8 / (180.0 + 2 \cdot (230.0 + 38.0))$$

$$= 16.815 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 12039.8 \cdot 70.0 / (180.0 \cdot (268.0) + 0.6667 \cdot (268.0)^2)$$

$$= 8.768 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{16.82^2 + 8.77^2}$$

$$= 18.964 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 18.96 / [(0.4 \cdot 2437) \cdot 2 \cdot 0.707]$$
$$= 1.376 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 12039.8 / (2 \cdot 130.0)$$
$$= 46.307 \text{ kgf/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 12039.8 \cdot 70.0 / (2 \cdot 268.0 \cdot 130.0)$$
$$= 12.095 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{46.31^2 + 12.1^2}$$
$$= 47.86 \text{ kgf/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 47.86 / [(0.4 \cdot 2437) \cdot 2 \cdot 0.707]$$
$$= 3.472 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.4722	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N33"

From: 30

Pressure for Reinforcement Calculations	P	46.553	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	20.0000	mm
Height of Beveled Transition	L'	28.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		38.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2000.0	mm
Centerline Distance to Tangent		2000.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N33"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.394	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 46.55 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.55)$
 $= 43.3205 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.55 \cdot 38.1 / (1407 \cdot 1.0 + 0.4 \cdot 46.55)$
 $= 1.2440 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 28.0 / 0.5773$
 $= 48.4974 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	174.2000	mm
Parallel to Vessel Wall	Rn+tn+t	87.1000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	65.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.1/1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.1/1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.946	NA	NA
Area in Shell	A1	6.361	NA	NA
Area in Nozzle Wall	A2	7.540	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	19.040	NA	NA
TOTAL AREA AVAILABLE	Atot	33.941	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (62.2 \cdot 43.3205 \cdot 1.0 + 2 \cdot 35.0 \cdot 43.3205 \cdot 1.0 \cdot (1 - 1.))$

= 26.946 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 112.0(1.0 \cdot 49.0 - 1.0 \cdot 43.321) - 2 \cdot 35.0 \\ &\quad (1.0 \cdot 49.0 - 1.0 \cdot 43.3205) \cdot (1 - 1.) \\ &= 6.361 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tnp)(tn - trn)fr2 \\ &= (2 \cdot 65.5)(7.0 - 1.24) \cdot 1. \\ &= 7.540 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi \cdot can / 0.707)^2 \cdot fr2 \\ &= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. \\ &= 1.000 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(65.5, 150.0, 20.0)) \cdot (38.0 - 10.0) \cdot 1. \\ &= 19.040 \text{ cm}^2 \text{ Includes Bevel Area of } 7.840 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2440 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 51.0965 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 51.0965 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.8, \max(51.0965, 4.5)] \\ &= 7.8000 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.244, 7.8) \\ &= 7.8000 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 587.5,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2930.3 kgf/cm ²	Passed
Occasional	: 93.0,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 391.0,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
Calculated Minimum Design Metal Temperature -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -86 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 46.55/104.01 = 0.448

Weld Size Calculations, Description: N33"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (26.9459 - 6.361 + 2 • 35.0 • 1. •
(1.0 • 49.0 - 43.3205))1407)
= 34560.52 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (26.5799 + 1. - 0.0 • 1.) • 1407
= 38809.19 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (26.5799 + 0.0 + 1. + (34.2994)) • 1407
= 87073.82 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (26.5799 + 0.0 + 1. + 0.0 + (34.2994)) • 1407
= 87073.82 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 76.2 • 10.0 • 0.49 • 1407
= 8253. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 34.6) • (38.0 - 3.0) • 0.7 • 1407
= 37474. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 76.2 • (52.0 - 3.0) • 0.74 • 1407
= 61072. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (8253 + 37474) = 45727 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (8253 + 0 + 61072 + 0) = 69325 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (8253 + 61072 + 0) = 69325 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 45726 kgf, must exceed W = 34560 kgf or W1 = 38809 kgf
 Path 2-2 = 69325 kgf, must exceed W = 34560 kgf or W2 = 87073 kgf
 Path 3-3 = 69325 kgf, must exceed W = 34560 kgf or W3 = 87073 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 47 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.2738 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 202.2737 mm

Input Echo, WRC107/537 Item 1, Description: N33" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5220.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	100.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.16	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.16	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

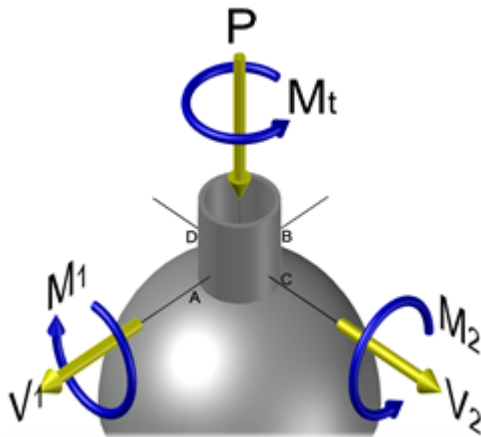
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	132.200	mm
Nozzle Thickness	Tn	38.000	mm
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13	kgf/cm ²
Design Internal Pressure	Dp	46.553	kgf/cm ²
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-611.6	kgf
Longitudinal Shear (SUS)	(V1) V1	407.7	kgf
Circumferential Shear (SUS)	(Vc) V2	407.7	kgf
Circumferential Moment (SUS)	(Mc) M1	91.3	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	91.3	kgf-m.
Torsional Moment (SUS)	Mt	130.5	kgf-m.

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 132.2 + 2 \cdot 1.65 \cdot \sqrt{(2637.5 (52.0 - 3.0))}$$

$$= 1318.537 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-611.6	kgf
Circumferential Shear	(VC)	V2	407.7	kgf
Longitudinal Shear	(VL)	V1	407.7	kgf
Circumferential Moment	(MC)	M1	91.3	kgf-m.
Longitudinal Moment	(ML)	M2	91.3	kgf-m.
Torsional Moment		MT	130.5	kgf-m.

Dimensionless Param: U = 0.18 TAU = 5.00 (1.39) RHO = 1.40

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.13840	(A,B,C,D)
$M(x) / P$	SP 2	0.16160	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.41475	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.66303	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.41475	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.66303	(A,B,C,D)
$N(y) * T / P$	SP 2	0.20395	(A,B,C,D)
$M(y) / P$	SP 2	0.09242	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.04134	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.50025	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.04134	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.50025	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Rad. Memb. P	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Rad. Bend. P	24.7	-24.7	24.7	-24.7	24.7	-24.7	24.7	-24.7
Rad. Memb. MC	0.0	0.0	0.0	0.0	-4.4	-4.4	4.4	4.4
Rad. Bend. MC	0.0	0.0	0.0	0.0	-42.1	42.1	42.1	-42.1
Rad. Memb. ML	-4.4	-4.4	4.4	4.4	0.0	0.0	0.0	0.0
Rad. Bend. ML	-42.1	42.1	42.1	-42.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-18.3	16.5	74.7	-58.9	-18.3	16.5	74.7	-58.9
Tang. Memb. P	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2
Tang. Bend. P	14.1	-14.1	14.1	-14.1	14.1	-14.1	14.1	-14.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-31.8	31.8	31.8	-31.8
Tang. Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-31.8	31.8	31.8	-31.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-12.9	22.4	51.5	-40.3	-12.9	22.4	51.5	-40.3
Shear VC	4.0	4.0	-4.0	-4.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-4.0	-4.0	4.0	4.0
Shear MT	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Tot. Shear	13.7	13.7	5.7	5.7	5.7	5.7	13.7	13.7
Str. Int.	29.5	33.5	76.0	60.5	21.9	25.9	81.1	66.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5
Rad. Pl (SUS)		-0.9	-0.9	7.9	7.9	-0.9	-0.9	7.9	7.9
Rad. Q (SUS)		-17.4	17.4	66.8	-66.8	-17.4	17.4	66.8	-66.8
Long. Pm (SUS)		1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5
Long. Pl (SUS)		4.8	4.8	5.6	5.6	4.8	4.8	5.6	5.6
Long. Q (SUS)		-17.6	17.6	45.9	-45.9	-17.6	17.6	45.9	-45.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.0	4.0	-4.0	-4.0	-4.0	-4.0	4.0	4.0
Shear Q (SUS)		9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Pm (SUS)		1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5	1241.5
Pm+Pl (SUS)		1248.4	1248.4	1252.5	1252.5	1248.4	1248.4	1252.5	1252.5
Pm+Pl+Q (Total)		1239.9	1275.0	1317.6	1202.9	1232.3	1267.4	1322.6	1208.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1241.54	1407.16	Passed
Pm+Pl (SUS)	1252.48	2110.74	Passed
Pm+Pl+Q (TOTAL)	1322.60	4221.48	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: N124"

From: 40

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		176.50	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	61.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	2860.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.551	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the

attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 46.41 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$
 $= 48.9180 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.41 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
 $= 9.9210 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 61.0 / 0.5773$
 $= 105.6552 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.0000	mm
Normal to Vessel Wall, Inward		20.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	287.443	NA	NA
Area in Shell	A1	6.358	NA	NA
Area in Nozzle Wall	A2	2.697	NA	NA
Area in Inward Nozzle	A3	172.497	NA	NA
Area in Welds A41+A42+A43		1.952	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	122.248	NA	NA
TOTAL AREA AVAILABLE	Atot	305.752	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (587.6 \cdot 48.918 \cdot 1.0 + 2 \cdot 72.0 \cdot 48.918 \cdot 1.0 \cdot (1 - 1.))$$
$$= 287.443 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$
$$= 587.6 (1.0 \cdot 50.0 - 1.0 \cdot 48.918) - 2 \cdot 72.0$$
$$(1.0 \cdot 50.0 - 1.0 \cdot 48.918) \cdot (1 - 1.)$$
$$= 6.358 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$
$$= (2 \cdot 125.0) (11.0 - 9.92) \cdot 1.$$
$$= 2.697 \text{ cm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, tl, 2.5 \cdot ti) \cdot fr2$$
$$= 2 \cdot 69.0 \cdot (125.0) \cdot 1.$$
$$= 172.497 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$
$$= 10.0^2 \cdot 1. + (9.7573)^2 \cdot 1.$$
$$= 1.952 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$
$$= (2 \cdot \min(125.0, 300.0, 70.0)) \cdot (75.0 - 14.0) \cdot 1.$$
$$= 122.248 \text{ cm}^2 \text{ Includes Bevel Area of } 36.849 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.9210 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 51.9180 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 51.9180 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[11.331, \max(51.918, 4.5)]$$
$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(12.921, 11.3312)$$
$$= 12.9210 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 825.1,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2692.7 kgf/cm ²	Passed
Occasional	: 608.3,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 165.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $46.41/52.01 = 0.892$

Weld Size Calculations, Description: N124"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm
Intermediate Calc. for Inward Weld	TminIns	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (287.4432 - 6.3581 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 50.0 - 48.918)) \cdot 1407) \\ &= 397723.56 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (124.9453 + 1.952 - 95.2053 \cdot 1.0) \cdot 1407 \\ &= 177224.78 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (124.9453 + 172.4969 + 1.952 + (71.9987)) \cdot 1407 \\ &= 522608.28 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (124.9453 + 172.4969 + 1.952 + 0.0 + (71.9987)) \cdot 1407 \\ &= 522608.28 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 66023. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 609.6 \cdot 9.7573 \cdot 0.49 \cdot 1407 \\ &= 64421. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 299.3) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 1407 \end{aligned}$$

= 666841. kgf

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (53.0 - 3.0) \cdot 0.74 \cdot 1407$$

$$= 498551. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66023 + 666841) = 732864 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66023 + 0 + 498551 + 64421) = 628995 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (66023 + 498551 + 64421) = 628995 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 732863 kgf, must exceed W = 397723 kgf or W1 = 177224 kgf
 Path 2-2 = 628994 kgf, must exceed W = 397723 kgf or W2 = 522608 kgf
 Path 3-3 = 628994 kgf, must exceed W = 397723 kgf or W3 = 522608 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 47 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3747 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 3213.0000 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 587.60 mm

Input Echo, WRC297 Item 1, Description: N124" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2902.000	mm
Shell Thickness	53.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.16	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.16	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	731.600	mm
Nozzle Thickness	75.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.



Design Internal Pressure	Dp	46.41	kgf/cm ²
Radial Load	P	-4587.18	kgf
Circumferential Shear	Vc	3058.09	kgf
Longitudinal Shear	Vl	3058.09	kgf
Circumferential Moment	Mc	5137.61	kgf-m.
Longitudinal Moment	Ml	5137.61	kgf-m.
Torsional Moment	Mt	7339.45	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.755) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 731.6 + 2 \cdot 1.65 \cdot \sqrt{(1479.0(53.0 - 3.0))}$
 $= 1628.994 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2958.000 mm
Nozzle Outside Diameter	(d) =	731.600 mm
Vessel Thickness used	(T) =	50.000 mm
Nozzle Thickness used	(t) =	72.000 mm
T / t	=	0.694
d / t	=	10.161

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.902$$

Nr/P	=	0.077
Mr/P	=	0.136
M0/P	=	0.041
N0/P	=	0.102
MrD/Mc	=	0.272
NrDL/Mc	=	0.116
M0d/Mc	=	0.089
N0DL/Mc	=	0.063
MrD/Ml	=	0.118
NrDL/Ml	=	0.070
M0D/Ml	=	0.040
N0DL/Ml	=	0.129

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	45	-45	45	-45
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-36	-36	36	36
Outplane Bending (Ml)	-67	67	67	-67
Normal Pressure Stress	1326	1373	1326	1373
Outplane Stress Summary	1287	1377	1494	1314

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	14	14	14	14
Inplane Bending (P)	149	-149	149	-149
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-19	-19	19	19
Inplane Bending (ML)	-198	198	198	-198
Inplane Pressure Stress	663	663	663	663
Inplane Stress Summary	608	706	1045	348

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	22	22	12	12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1287	1378	1495	1314

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	45	-45	45	-45
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-150	150	150	-150
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	663	663	663	663
Outplane Stress Summary	559	769	895	503

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	14	14	14	14
Inplane Bending (P)	149	-149	149	-149
Inplane Membrane (Mc)	-32	-32	32	32
Inplane Bending (Mc)	-458	458	458	-458
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1326	1373	1326	1373
Inplane Stress Summary	999	1663	1981	811

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	17	17	17	17
Shear Stress Summary	12	12	22	22

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 999 1663 1981 813

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses in the	Au	Al	Bu	B1
hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-36	-36	36	36
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	202	258	202	258
Hoop Stress Summary	184	240	257	312

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to	Au	Al	Bu	B1
pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	51	-51	51	-51
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-67	67	67	-67
Axial Pressure Stress	101	101	101	101
Axial Stress Summary	65	96	246	8

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	8	8

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	187	242	261	313

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu	Cl	Du	D1
hoop direction)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	202	258	202	258
Hoop Stress Summary	203	259	238	294

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu	Cl	Du	D1
pipe cross-section)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	51	-51	51	-51

Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-173	173	173	-173
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	101	101	101	101
Axial Stress Summary	-40	203	352	-98

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	8	8	15	15

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	244	260	354	394

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1326	1373	1326	1373	1326	1373	1326	1373
Circ. Pl (SUS)	-18	-18	54	54	-18	-18	46	46
Circ. Q (SUS)	-22	22	112	-112	-309	309	607	-607
Long. Pm (SUS)	663	663	663	663	663	663	663	663
Long. Pl (SUS)	-5	-5	33	33	1	1	35	35
Long. Q (SUS)	-49	49	347	-347	-105	105	195	-195
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1326	1373	1326	1373	1326	1373	1326	1373
Pm+Pl (SUS)	1308	1355	1380	1427	1308	1355	1372	1419
Pm+Pl+Q (Total)	1286	1377	1492	1315	999	1664	1979	813

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1373.00	1407.16	Passed
Pm+Pl (SUS)	1427.03	2110.74	Passed
Pm+Pl+Q (TOTAL)	1979.45	4221.48	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	202	258	202	258	202	258	202	258
Circ. Pl (SUS)	-18	-18	54	54	1	1	35	35
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	101	101	101	101	101	101	101	101
Long. Pl (SUS)	-19	-19	25	25	-19	-19	25	25
Long. Q (SUS)	-16	16	118	-118	-122	122	224	-224
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	202.0	258.0	202.0	258.0	202.0	258.0	202.0	258.0
Pm+Pl (SUS)	184.1	240.1	256.1	312.0	203.1	259.1	237.1	293.1
Pm+Pl+Q (Total)	185.9	241.6	260.8	312.3	243.7	260.4	352.0	392.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	258.00	1407.13	Passed
Pm+Pl (SUS)	312.05	2110.70	Passed
Pm+Pl+Q (TOTAL)	392.15	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N41.5"

From: 40

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		140.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N41.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.630	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

$$= 46.41 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$$
$$= 48.9180 \text{ mm}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $22.05(\exp([46.41/(1407.13 \cdot 1.0)] - 1))$
= 0.7393 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	170.1040	mm
Parallel to Vessel Wall	Rn+tn+t	85.0520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N41.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.7393	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9180	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9180	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.52, max(51.918, 4.5)]
= 7.5200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.7393, 7.52)
= 7.5200 mm

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 190.8,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3327.0 kgf/cm ²	Passed
Occasional	: 30.4,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 168.3,	Allowable	: 985.0 kgf/cm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Flange Material Test Exemption Temperature per Specification: -45.6 C

Weld Size Calculations, Description: N41.5"



Intermediate Calc. for nozzle/shell Welds Tmin 13.0020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4234 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.4234 mm

Input Echo, WRC297 Item 1, Description: N41.5" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2902.000	mm
Shell Thickness	53.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.16	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.16	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	38.100	mm
Nozzle Thickness	16.0020	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	46.41	kgf/cm ²
Radial Load	P	-382.26	kgf
Circumferential Shear	Vc	254.84	kgf
Longitudinal Shear	Vl	254.84	kgf
Circumferential Moment	Mc	35.67	kgf-m.
Longitudinal Moment	Ml	35.67	kgf-m.
Torsional Moment	Mt	50.96	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (1.323) is < 5
 check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.381) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 70.104 + 2 \cdot 1.65 \cdot \sqrt{(1479.0(53.0 - 3.0))}$$

$$= 967.498 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2958.000 mm



Nozzle Outside Diameter (d) = 70.104 mm
 Vessel Thickness used (T) = 50.000 mm
 Nozzle Thickness used (t) = 13.002 mm
 T / t = 3.846
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.182$$

Nr/P = 0.254
 Mr/P = 0.203
 M0/P = 0.290
 N0/P = 0.206
 MrD/Mc = 0.158
 NrDL/Mc = 0.174
 M0d/Mc = 0.258
 N0DL/Mc = -0.133
 MrD/Ml = 0.158
 NrDL/Ml = 0.179
 M0D/Ml = 0.256
 N0DL/Ml = -0.146

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	2	2	-2	-2
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	1326	1373	1326	1373
Outplane Stress Summary	1328	1383	1384	1315

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-19	19	19	-19
Inplane Pressure Stress	663	663	663	663
Inplane Stress Summary	662	664	708	632

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	13	13	13	13
Shear Stress Summary	17	17	8	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	1328	1384	1384	1315

Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	663	663	663	663
Outplane Stress Summary	664	673	721	605

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1326	1373	1326	1373
Inplane Stress Summary	1326	1374	1371	1342

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	13	13	13	13
Shear Stress Summary	8	8	17	17

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1326	1374	1372	1342

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	2	2	-2	-2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	72	128	72	128
Hoop Stress Summary	78	134	73	128

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	16	16	16	16
Axial Bending (P)	102	-102	102	-102
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-125	-125	125	125
Axial Bending (ML)	-124	124	124	-124
Axial Pressure Stress	36	36	36	36

Axial Stress Summary	-93	-50	404	-49
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	17	17	-17	-17
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	50	50	50	50

Shear Stress Summary	68	68	32	32
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	220	230	407	189
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	72	128	72	128

Hoop Stress Summary	78	134	73	129
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	16	16	16	16
Axial Bending (P)	102	-102	102	-102
Axial Membrane (Mc)	-125	-125	125	125
Axial Bending (Mc)	-129	129	129	-129
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	36	36	36	36

Axial Stress Summary	-98	-45	409	-53
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-17	-17	17	17
Shear due to Torsion	50	50	50	50

Shear Stress Summary	32	32	68	68
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	189	191	422	228
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Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1326	1373	1326	1373	1326	1373	1326	1373
Circ. Pl (SUS)	5	5	1	1	0	0	6	6
Circ. Q (SUS)	-5	5	57	-57	-1	1	37	-37
Long. Pm (SUS)	663	663	663	663	663	663	663	663
Long. Pl (SUS)	0	0	6	6	5	5	1	1
Long. Q (SUS)	-1	1	37	-37	-5	5	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	1326	1373	1326	1373	1326	1373	1326	1373
Pm+Pl (SUS)	1331	1378	1327	1374	1326	1373	1332	1379
Pm+Pl+Q (Total)	1326	1383	1384	1317	1325	1374	1369	1342

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1373.00	1407.16	Passed
Pm+Pl (SUS)	1379.02	2110.74	Passed
Pm+Pl+Q (TOTAL)	1384.12	4221.48	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	72	128	72	128	72	128	72	128
Circ. Pl (SUS)	5	5	1	1	5	5	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	36	36	36	36	36	36	36	36
Long. Pl (SUS)	-109	-109	141	141	-109	-109	141	141
Long. Q (SUS)	-22	22	226	-226	-27	27	231	-231
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	17	17	-17	-17	-17	-17	17	17
Shear Q (SUS)	50	50	50	50	50	50	50	50
Pm (SUS)	72.0	128.0	72.0	128.0	72.0	128.0	72.0	128.0
Pm+Pl (SUS)	153.8	208.8	179.7	182.4	153.8	208.8	179.7	182.4
Pm+Pl+Q (Total)	218.0	227.6	406.3	189.8	188.9	190.8	420.9	226.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	1407.13	Passed
Pm+Pl (SUS)	208.79	2110.70	Passed

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Pm+Pl+Q (TOTAL)	420.90	4221.40	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K12"

From: 40

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		108.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

$$= 46.41 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$$
$$= 48.9180 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $46.41 \cdot 28.4 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$
= 0.9555 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+t	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K12".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9555	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9180	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9180	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.918, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9555, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Flange Material Test Exemption Temperature per Specification: -45.6 C

Weld Size Calculations, Description: K12"

Intermediate Calc. for nozzle/shell Welds	Tmin	13.7640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.6128 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6127 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 92.16 mm

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Input, Nozzle Desc: K22"
From: 40

Pressure for Reinforcement Calculations	P	46.465	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

$$= 46.47 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.47)$$
$$= 48.9824 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $46.47 \cdot 28.4 / (1407 \cdot 1.0 - 0.6 \cdot 46.47)$
= 0.9568 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+t	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K22".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9568	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9824	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9824	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.9824, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9568, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Flange Material Test Exemption Temperature per Specification: -45.6 C

Weld Size Calculations, Description: K22"

Intermediate Calc. for nozzle/shell Welds	Tmin	13.7640	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.6128 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6127 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 92.16 mm

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Input, Nozzle Desc: K32"

From: 40

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		85.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

$$= 46.41 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$$
$$= 48.9180 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $46.41 \cdot 28.4 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$
= 0.9555 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+t	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K32".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9555	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9180	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9180	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.918, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9555, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Flange Material Test Exemption Temperature per Specification: -45.6 C

Weld Size Calculations, Description: K32"

Intermediate Calc. for nozzle/shell Welds	Tmin	13.7640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.6128 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6127 mm



Input, Nozzle Desc: K42"

From: 40

Pressure for Reinforcement Calculations	P	46.465	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

$$= 46.47 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.47)$$
$$= 48.9824 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $46.47 \cdot 28.4 / (1407 \cdot 1.0 - 0.6 \cdot 46.47)$
= 0.9568 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+t	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K42".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9568	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9824	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9824	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.9824, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9568, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Flange Material Test Exemption Temperature per Specification: -45.6 C

Weld Size Calculations, Description: K42"

Intermediate Calc. for nozzle/shell Welds	Tmin	13.7640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6128 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6127 mm



Input, Nozzle Desc: M124"
From: 40

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2902.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		100.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	116.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		130.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.551	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:



For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 46.41 \cdot 1454.0 / (1407 \cdot 1.0 - 0.6 \cdot 46.41)$
 $= 48.9180 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.41 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
 $= 9.9210 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 116.0 / 0.5773$
 $= 200.9180 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	287.444	NA	NA
Area in Shell	A1	6.358	NA	NA
Area in Nozzle Wall	A2	2.697	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	283.745	NA	NA
TOTAL AREA AVAILABLE	Atot	293.800	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (587.6 \cdot 48.918 \cdot 1.0 + 2 \cdot 127.0 \cdot 48.918 \cdot 1.0 \cdot (1 - 1.))$$
$$= 287.444 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$
$$= 587.6(1.0 \cdot 50.0 - 1.0 \cdot 48.918) - 2 \cdot 127.0$$
$$(1.0 \cdot 50.0 - 1.0 \cdot 48.918) \cdot (1 - 1.)$$
$$= 6.358 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$
$$= (2 \cdot 125.0)(11.0 - 9.92) \cdot 1.$$
$$= 2.697 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$
$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$
$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$
$$= (2 \cdot \min(125.0, 300.0, 100.0)) \cdot (130.0 - 14.0) \cdot 1.$$
$$= 283.745 \text{ cm}^2 \text{ Includes Bevel Area of } 51.749 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 12.9210 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$
$$= 12.9210 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 46.41/52.01 = 0.892$$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv) \\ &= \max(0, (287.4442 - 6.3581 + 2 \cdot 127.0 \cdot 1. \cdot \\ &\quad (1.0 \cdot 50.0 - 48.918)) \cdot 1407) \\ &= 399399.81 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (286.4425 + 1. - 0.0 \cdot 1.) \cdot 1407 \\ &= 404476.81 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (286.4425 + 0.0 + 1. + (126.9977)) \cdot 1407 \\ &= 583182.62 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (286.4425 + 0.0 + 1. + 0.0 + (126.9977)) \cdot 1407 \\ &= 583182.62 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407 \\ &= 66023. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 299.3) \cdot (130.0 - 3.0) \cdot 0.7 \cdot 1407 \\ &= 1176232. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 609.6 \cdot (53.0 - 3.0) \cdot 0.74 \cdot 1407 \\ &= 498551. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66023 + 1176233) = 1242256 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (66023 + 0 + 498551 + 0) = 564574 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (66023 + 498551 + 0) = 564574 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

$$\begin{aligned} \text{Path 1-1} &= 1242255 \text{ kgf, must exceed } W = 399399 \text{ kgf} & \text{or } W1 &= 404476 \text{ kgf} \\ \text{Path 2-2} &= 564573 \text{ kgf, must exceed } W = 399399 \text{ kgf} & \text{or } W2 &= 583182 \text{ kgf} \\ \text{Path 3-3} &= 564573 \text{ kgf, must exceed } W = 399399 \text{ kgf} & \text{or } W3 &= 583182 \text{ kgf} \end{aligned}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 47 kgf/cm²

The Drop for this Nozzle is : 32.3747 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 385.3746 mm

Input, Nozzle Desc: N224"

From: 50

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	96.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N224"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.551	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:



For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 46.41 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.41)$
 $= 43.1829$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.41 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
 $= 9.9210$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 96.0 / 0.5773$
 $= 166.2769$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.1/1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.1/1407.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	253.744	NA	NA
Area in Shell	A1	34.181	NA	NA
Area in Nozzle Wall	A2	2.643	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	230.133	NA	NA
TOTAL AREA AVAILABLE	Atot	267.958	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)

$$= (587.6 \cdot 43.1829 \cdot 1.0 + 2 \cdot 107.0 \cdot 43.1829 \cdot 1.0 \cdot (1 - 1.))$$
$$= 253.744 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$
$$= 587.6 (1.0 \cdot 49.0 - 1.0 \cdot 43.183) - 2 \cdot 107.0$$
$$(1.0 \cdot 49.0 - 1.0 \cdot 43.1829) \cdot (1 - 1.)$$
$$= 34.181 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$
$$= (2 \cdot 122.5) (11.0 - 9.92) \cdot 1.$$
$$= 2.643 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$
$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$
$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$
$$= (2 \cdot \min(122.5, 500.0, 100.0)) \cdot (110.0 - 14.0) \cdot 1.$$
$$= 230.133 \text{ cm}^2 \text{ Includes Bevel Area of } 38.137 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 12.9210 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 50.9436 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 50.9436 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[11.331, \max(50.9436, 4.5)]$$
$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(12.921, 11.3312)$$
$$= 12.9210 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 825.1,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2692.7 kgf/cm ²	Passed
Occasional	: 608.3,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 165.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 46.41/52.01 = 0.892

Weld Size Calculations, Description: N224"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (253.7442 - 34.1813 + 2 \cdot 107.0 \cdot 1. \cdot (1.0 \cdot 49.0 - 43.1829)) \cdot 1407)$
 $= 326476.53 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (232.7769 + 1. - 0.0 \cdot 1.) \cdot 1407$
 $= 328960.91 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (232.7769 + 0.0 + 1. + (104.8581)) \cdot 1407$
 $= 476512.81 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (232.7769 + 0.0 + 1. + 0.0 + (104.8581)) \cdot 1407$
 $= 476512.81 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 66023. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 299.3) \cdot (110.0 - 3.0) \cdot 0.7 \cdot 1407$
 $= 990999. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (52.0 - 3.0) \cdot 0.74 \cdot 1407$
 $= 488580. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (66023 + 990999) = 1057022 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (66023 + 0 + 488580 + 0) = 554603 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (66023 + 488580 + 0) = 554603 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1057022 kgf, must exceed W = 326476 kgf or W1 = 328960 kgf
 Path 2-2 = 554602 kgf, must exceed W = 326476 kgf or W2 = 476512 kgf
 Path 3-3 = 554602 kgf, must exceed W = 326476 kgf or W3 = 476512 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 47 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 17.5766 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 569.5765 mm

Input Echo, WRC107/537 Item 1, Description: N224" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5220.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	100.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.16	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.16	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	801.600	mm
Nozzle Thickness	Tn	110.000	mm
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13	kgf/cm ²
Design Internal Pressure	Dp	46.405	kgf/cm ²
Include Pressure Thrust		No	

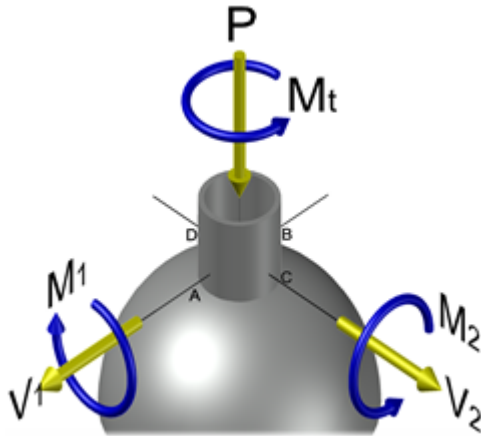
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-4587.1	kgf
Longitudinal Shear (SUS)	(V1) V1	3058.1	kgf
Circumferential Shear (SUS)	(Vc) V2	3058.1	kgf
Circumferential Moment (SUS)	(Mc) M1	5137.6	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	5137.6	kgf-m.
Torsional Moment (SUS)	Mt	7339.4	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 801.6 + 2 \cdot 1.65 \cdot \sqrt{(2637.5 (52.0 - 3.0))}$$

$$= 1987.937 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-4587.1	kgf
Circumferential Shear	(VC) V2	3058.1	kgf
Longitudinal Shear	(VL) V1	3058.1	kgf
Circumferential Moment	(MC) M1	5137.6	kgf-m.
Longitudinal Moment	(ML) M2	5137.6	kgf-m.
Torsional Moment	MT	7339.4	kgf-m.

Dimensionless Param: U = 1.11 TAU = 5.00 (3.25) RHO = 0.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.03626	(A,B,C,D)
M(x) / P	SP 1	0.03750	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.04110	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.06412	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.04110	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.06412	(A,B,C,D)
N(y) * T / P	SP 1	0.05768	(A,B,C,D)
M(y) / P	SP 1	0.01074	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.05685	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.02023	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.05685	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.02023	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Rad. Bend.	P	43.0	-43.0	43.0	-43.0	43.0	-43.0	43.0	-43.0
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-24.5	-24.5	24.5	24.5



Rad. Bend. MC	0.0	0.0	0.0	0.0	-229.0	229.0	229.0	-229.0
Rad. Memb. ML	-24.5	-24.5	24.5	24.5	0.0	0.0	0.0	0.0
Rad. Bend. ML	-229.0	229.0	229.0	-229.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-203.5	168.5	303.4	-240.6	-203.5	168.5	303.4	-240.6
Tang. Memb. P	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0
Tang. Bend. P	12.3	-12.3	12.3	-12.3	12.3	-12.3	12.3	-12.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-33.8	-33.8	33.8	33.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-72.3	72.3	72.3	-72.3
Tang. Memb. ML	-33.8	-33.8	33.8	33.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-72.3	72.3	72.3	-72.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-82.8	37.1	129.4	-39.7	-82.8	37.1	129.4	-39.7
Shear VC	5.0	5.0	-5.0	-5.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-5.0	-5.0	5.0	5.0
Shear MT	14.8	14.8	14.8	14.8	14.8	14.8	14.8	14.8
Tot. Shear	19.8	19.8	9.9	9.9	9.9	9.9	19.8	19.8
Str. Int.	206.7	171.4	303.9	241.1	204.3	169.2	305.6	242.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6
Rad. Pl (SUS)		-17.5	-17.5	31.4	31.4	-17.5	-17.5	31.4	31.4
Rad. Q (SUS)		-186.0	186.0	272.0	-272.0	-186.0	186.0	272.0	-272.0
Long. Pm (SUS)		1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6
Long. Pl (SUS)		-22.8	-22.8	44.9	44.9	-22.8	-22.8	44.9	44.9
Long. Q (SUS)		-59.9	59.9	84.6	-84.6	-59.9	59.9	84.6	-84.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		5.0	5.0	-5.0	-5.0	-5.0	-5.0	5.0	5.0
Shear Q (SUS)		14.8	14.8	14.8	14.8	14.8	14.8	14.8	14.8
Pm (SUS)		1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6	1237.6
Pm+Pl (SUS)		1223.0	1223.0	1284.1	1284.1	1223.0	1223.0	1284.1	1284.1
Pm+Pl+Q (Total)		1158.0	1409.0	1541.5	1198.4	1155.7	1406.8	1543.2	1199.8

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1237.61	1407.16	Passed
Pm+Pl (SUS)	1284.09	2110.74	Passed
Pm+Pl+Q (TOTAL)	1543.19	4221.48	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K52"

From: 50

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	10.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $46.41 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.41)$
= 43.1829 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $46.41 \cdot 25.4 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
= 0.8268 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20.0 / 0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	152.7548	mm
Parallel to Vessel Wall	Rn+tn+t	76.3774	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K52".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $46.41 / 52.01 = 0.892$



Weld Size Calculations, Description: K52"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $46.41 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.41)$
= 43.1829 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $46.41 \cdot 25.4 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
= 0.8268 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20.0/0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	148.8000 mm
Parallel to Vessel Wall	Rn+tn+t	74.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K52".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.8268 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 50.9436 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 50.9436 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[6.42, max(50.9436, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.8268, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K52"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 10.4966 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 366.5531 mm

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Input, Nozzle Desc: K62"

From: 50

Pressure for Reinforcement Calculations	P	46.405	kgf/cm ²
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.16	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.16	kgf/cm ²
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	10.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $46.41 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.41)$
= 43.1829 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $46.41 \cdot 25.4 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
= 0.8268 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $20.0 / 0.5773$
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	152.7548	mm
Parallel to Vessel Wall	Rn+tn+t	76.3774	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K62".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $46.41 / 52.01 = 0.892$

Weld Size Calculations, Description: K62"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 = $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 = $46.41 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 1407.16 \cdot 1.0 - 0.2 \cdot 46.41)$
 = 43.1829 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $46.41 \cdot 25.4 / (1407 \cdot 1.0 + 0.4 \cdot 46.41)$
 = 0.8268 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 20.0 / 0.5773
 = 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	148.8000 mm
Parallel to Vessel Wall	Rn+tn+t	74.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K62".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.8268 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 50.9436 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 50.9436 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[6.42, max(50.9436, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.8268, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: K62"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 10.4966 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 366.5531 mm

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